Robust Strictly Positive Real Synthesis for Polynomial Segments: Formal Verification of the Weak SPR Region Intersection Method

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Abstract

This paper presents a Lean formalization of the robust strictly positive real (SPR) synthesis theorem for polynomial segments of arbitrary positive order, previously established by Yu, Wang, and Ackermann. The proof follows the weak SPR (WSPR) region intersection strategy of Wang and Yu: obtain a common numerator of one lower degree and then raise its degree by a small perturbation. For monic real Hurwitz denominators of degree n, stability of every convex combination is equivalent to the existence of a common monic numerator of degree exactly $n-1$ satisfying strict frequency positivity. The same numerator works on the whole segment. The intersection proof uses frequency compactification of degree $2n-2$, convex separation, and a finite certificate at a transition from feasibility to failure. Analytic properties of weakly positive real quotients determine the degree and root structure of the supporting polynomial. A signed residue reflection identity then contradicts the contact count, including contacts at infinity. A common perturbation and positive normalization give the corresponding monic degree-n SPR numerator. The contribution of this work is a machine-checked proof of this established synthesis result, with a uniform algebraic treatment of the region-intersection step.

Keywords: robust SPR synthesis; weak SPR region; polynomial segment; convex separation; residue reflection; Lean.

1 Introduction

Robust strictly positive real synthesis asks for one numerator whose quotient with every denominator in an uncertain family has positive real part on the imaginary axis. For a polynomial segment, the denominators are the convex combinations of two fixed real polynomials. The problem is to choose one numerator before the segment parameter and the frequency are specified. This quantifier order is essential: a separate numerator for each denominator does not solve the robust synthesis problem.

Wang and Yu introduced weak SPR regions in numerator coefficient space and developed a synthesis method based on their intersections [1]. In their terminology, an SPR quotient has numerator and denominator of the same degree, whereas a WSPR quotient has a numerator of one lower degree. Both require a Hurwitz denominator and strictly positive real part at every real frequency. A common point in the WSPR regions supplies a lower-degree numerator, which can subsequently be perturbed to obtain an SPR numerator.

Yu, Wang, and Ackermann established the arbitrary-order polynomial-segment synthesis criterion in [2,3]. The geometric proof constructs lower-degree candidates through elliptic regions and separating hyperplanes, followed by perturbation. These works establish the synthesis theorem and the two-stage logic used here. The present contribution is a formal verification of that logic, with a uniform algebraic realization of the region-intersection argument.

The accompanying file `SPR/simplified/WSPR.lean` first proves existence of a monic numerator of degree exactly $n-1$. Its proof searches in the n-dimensional space of polynomials of degree at most $n-1$. Frequency compactification turns positivity into strict separation from a compact convex hull. A hypothetical failure of intersection produces a nonzero supporting polynomial and finitely many weighted contacts. Signed frequency moments and a residue reflection identity exclude this certificate. Finally, a small perturbation gives a monic numerator of degree n. All orders are handled by the same argument, with $n = 1$ treated directly.

The organizing object of this proof is the lower-degree WSPR region. Its compactification has degree $2n-2$, and a contact at infinity removes one usable moment while also decreasing the contact-count bound. The reflection argument accommodates both cases, including the limiting case where its polynomial degree bound is attained. This gives a precise verification of the intersection step without identifying the formal proof with every geometric construction in the earlier literature.

2 Definitions and main results

Let $\mathcal{H}_n$ be the set of monic real polynomials of degree n with no zero in the closed right half-plane. Thus

$$p \in \mathcal{H}_n \quad \Longleftrightarrow \quad p \text{ is monic}, \quad \deg p = n, \quad p(z) \neq 0 \ \text{ whenever } \operatorname{Re} z \geq 0. \tag{2.1}$$

Fix monic real polynomials a, b of degree n, and put

$$p_\lambda = (1-\lambda)a + \lambda b, \qquad 0 \leq \lambda \leq 1. \tag{2.2}$$

The whole segment has degree n because its leading coefficient is always one.

Definition 2.1 (SPR and WSPR)**.** For a real Hurwitz polynomial p of degree n, a quotient c/p is WSPR if $\deg c = n-1$ and

$$\operatorname{Re} \frac{c(i\omega)}{p(i\omega)} > 0 \qquad (\omega \in \mathbb{R}). \tag{2.3}$$

It is SPR if $\deg c = n$ and (2.3) holds. Throughout the synthesis statements, both denominators and the resulting common numerator are monic.

This is the degree-based convention of [1, Definitions 1 and 2]. In particular, WSPR retains the strict inequality in (2.3). A nonnegative inequality used for a supporting polynomial later in the proof describes a boundary condition, not the WSPR definition. No additional positive limit at infinity is imposed on a WSPR quotient; its limit is zero. The formal predicate named `SPR` records denominator stability and (2.3); the degree and monicity constraints occur separately in the theorem statements.

For $p \in \mathcal{H}_n$, define the normalized weak SPR region by

$$\Omega_p^W = \{c \in \mathbb{R}[s] : c \text{ monic}, \ \deg c = n-1, \ c/p \in \mathrm{WSPR}\}. \tag{2.4}$$

Writing $c = s^{n-1} + x_1 s^{n-2} + \cdots + x_{n-1}$ identifies this region with a subset of $\mathbb{R}^{n-1}$. The strict frequency inequalities are affine in these coefficients, so the region is convex. An intersection $\Omega_a^W \cap \Omega_b^W$ consists of common normalized WSPR numerators.

Theorem 2.2 (Weak SPR region intersection)**.** *Let $n \geq 1$, and let a, b be monic real polynomials of degree n. The following statements are equivalent:*

1. *$p_\lambda \in \mathcal{H}_n$ for every $\lambda \in [0,1]$;*
2. *$a, b \in \mathcal{H}_n$ and $\Omega_a^W \cap \Omega_b^W \neq \varnothing$;*
3. *there is a single monic real polynomial c of degree exactly $n-1$ such that c/p_λ is WSPR for every $\lambda \in [0,1]$.*

The numerator in the third statement can therefore be chosen to satisfy

$$\operatorname{Re} \frac{c(i\omega)}{p_\lambda(i\omega)} > 0 \qquad (\lambda \in [0,1],\ \omega \in \mathbb{R}). \tag{2.5}$$

Theorem 2.3 (Robust SPR synthesis)**.** *Let $n \geq 0$, and let a, b be monic real polynomials of degree n. The segment (2.2) is Hurwitz stable if and only if there exists a monic real polynomial β of degree exactly n for which β/a and β/b are SPR. The same β then makes β/p_λ SPR for every $\lambda \in [0,1]$.*

For $n = 0$, all three monic polynomials are 1. For positive n, Theorem 2.3 will follow from Theorem 2.2 by a common perturbation.

3 Frequency geometry and compactification

3.1 The cross polynomial

For $c, p \in \mathbb{R}[s]$, let

$$R_{c,p}(\omega) = \operatorname{Re}\big(c(i\omega)\overline{p(i\omega)}\big). \tag{3.1}$$

This is a real polynomial in ω. If $p(i\omega) \neq 0$, then

$$\operatorname{Re} \frac{c(i\omega)}{p(i\omega)} = \frac{R_{c,p}(\omega)}{|p(i\omega)|^2}. \tag{3.2}$$

Real coefficients imply $R_{c,p}(-\omega) = R_{c,p}(\omega)$. If $\deg c \leq n-1$ and $\deg p \leq n$, evenness improves the product degree bound to

$$\deg R_{c,p} \leq 2n-2. \tag{3.3}$$

For $n \geq 2$, writing coefficients by ascending powers gives

$$L_p(c) := [\omega^{2n-2}]R_{c,p} = c_{n-1}p_{n-1} - c_{n-2}p_n. \tag{3.4}$$

Here $p_n = 1$ for a monic denominator. The cross polynomial is linear in c and affine in p; in particular,

$$R_{c,p_\lambda} = (1-\lambda)R_{c,a} + \lambda R_{c,b}. \tag{3.5}$$

Consequently, positivity for two endpoints implies positivity for every intermediate denominator. The remaining issue in the converse direction is preservation of denominator stability.

3.2 A compact sufficient system

For $n \geq 2$, set $d = 2n-2$. For $q(\omega) = \sum_{k=0}^{d} q_k \omega^k$, define

$$q^{[d]}(x,y) = \sum_{k=0}^{d} q_k x^k y^{d-k}, \qquad B_p(c,u) = R_{c,p}^{[d]}(u, 1-|u|), \quad -1 \leq u \leq 1. \tag{3.6}$$

For $|u| < 1$ this becomes

$$B_p(c,u) = (1-|u|)^d R_{c,p}\left(\frac{u}{1-|u|}\right), \tag{3.7}$$

whereas $B_p(c,1) = B_p(c,-1) = L_p(c)$. For a stable denominator, strict positivity of $B_p(c,\cdot)$ on $[-1,1]$ is thus equivalent to (2.3) together with $L_p(c) > 0$.

The coefficient condition $L_p(c) > 0$ is an additional sufficient condition used in the proof. It is not part of Definition 2.1. For example, let

$$p = s^3 + s^2 + 2s + 1, \qquad c = s^2 + s + 1.$$

Both polynomials are Hurwitz stable, and $R_{c,p}(\omega) = 1$. Thus c/p is WSPR although $L_p(c) = 0$. We will prove existence of a common numerator satisfying the stronger compact system, which is enough to establish intersection of the WSPR regions.

Let $V_n = \{c \in \mathbb{R}[s] : \deg c \le n-1\} \simeq \mathbb{R}^n$, with the zero polynomial included. For $0 \le k \le n-1$, define

$$\Phi_p(u)_k = B_p(s^k, u). \tag{3.8}$$

If $v = (c_0, \dots, c_{n-1})$, then $B_p(c,u) = v \cdot \Phi_p(u)$. The set

$$K(a,b) = \Phi_a([-1,1]) \cup \Phi_b([-1,1]), \qquad C(a,b) = \operatorname{conv} K(a,b) \tag{3.9}$$

is compact, as is its convex hull in finite dimension. Strict separation gives

$$0 \notin C(a,b) \iff \exists c \in V_n : \; B_a(c,u) > 0 \text{ and } B_b(c,u) > 0 \text{ for every } u \in [-1,1]. \tag{3.10}$$

This equivalence concerns compact feasibility. Exact degree and positive leading coefficient will be derived after a separator has been obtained.

3.3 A feasible starting point

For a single $p \in \mathcal{H}_n$, $n \ge 1$, take $c = p'/n$. If $\rho_1, \dots, \rho_n$ are its roots, counted with multiplicity, then

$$\operatorname{Re}\frac{p'(i\omega)}{np(i\omega)} = \frac{1}{n}\sum_{j=1}^{n}\frac{-\operatorname{Re}\rho_j}{|i\omega - \rho_j|^2} > 0. \tag{3.11}$$

The polynomial p'/n is monic of degree $n-1$. For $n \ge 2$, (3.4) yields

$$L_p(p'/n) = \frac{p_{n-1}}{n} > 0, \tag{3.12}$$

because $p_{n-1} = -\sum_j \operatorname{Re}\rho_j > 0$. If $n = 1$, write $p = s + \alpha$, $\alpha > 0$; then $c = 1$ and $B_p(1,u) = \alpha$. Hence the compact system is feasible at the coincident pair (p,p) in every positive order.

4 A critical support and its analytic properties

4.1 From failed intersection to contact

Assume $n \ge 2$ and the entire segment is stable, but the compact system for (a,b) is infeasible. For $t \in [0,1]$, consider the pair (a,p_t) and define the continuous quality function

$$\gamma(t) = \max_{\|v\|=1} \min_{\substack{p \in \{a,p_t\} \\ u \in [-1,1]}} v \cdot \Phi_p(u). \tag{4.1}$$

The inner minimum and outer maximum are attained on compact sets. Their continuous dependence on t follows from continuity of the feature map. By (3.11), $\gamma(0) > 0$; infeasibility implies $\gamma(1) \le 0$. There is therefore a transition parameter τ with $\gamma(\tau) = 0$.

An attaining vector of norm one determines a nonzero polynomial $h \in V_n$ satisfying

$$B_a(h,u) \ge 0, \qquad B_{p_\tau}(h,u) \ge 0 \qquad (-1 \le u \le 1). \tag{4.2}$$

At the same parameter, $0 \in C(a, p_\tau)$. A finite convex representation, with zero weights removed, gives points $z_j = \Phi_{d_j}(u_j)$ and weights $\mu_j > 0$, where $d_j \in \{a, p_\tau\}$, such that

$$\sum_j \mu_j = 1, \qquad \sum_j \mu_j z_j = 0, \qquad B_{d_j}(h,u_j) = 0. \tag{4.3}$$

The contact equalities follow by pairing the balance identity with the nonnegative support h. The unit-norm condition ensures that this support cannot vanish. The new denominator pair also has a stable segment, since

$$(1-v)a + vp_\tau = p_{v\tau} \qquad (0 \le v \le 1).$$

For the remainder of the certificate argument, relabel this pair as (a, b).

4.2 Degree and roots of a weakly positive support

Lemma 4.1. *Let $p \in \mathcal{H}_n$, $n \ge 1$, and let $0 \ne h \in V_n$. If*

$$\operatorname{Re} \frac{h(i\omega)}{p(i\omega)} \ge 0 \qquad (\omega \in \mathbb{R}), \tag{4.4}$$

then $\deg h = n-1$, all roots of h lie in the closed left half-plane, and every imaginary-axis root of h is simple. Moreover, $R_{h,p}$ is a nonzero nonnegative real polynomial.

Proof. Put $F = h/p$. The denominator is nonzero on the closed right half-plane, and $F(s) \to 0$ as $s \to \infty$. The Cayley map $s = (1+z)/(1-z)$ transports F to a function analytic on a neighborhood of the closed disk, with a removable value 0 at $z = 1$. The minimum principle for its real part extends (4.4) to the whole right half-plane.

At a zero s_0, write $F(s_0 + \zeta) = A\zeta^m + O(\zeta^{m+1})$, with $A \ne 0$. An interior zero would allow directions making the leading real part negative. At a boundary zero, the permitted directions fill a half-circle; if $m \ge 2$, they again include a direction with negative leading real part. Hence there are no open right-half-plane roots, and boundary zeros are simple. Applying the same boundary argument at $z = 0$ to $F(1/z)$ shows that its zero there has order at most one. That order is $n - \deg h$, so $\deg h = n - 1$.

Equation (3.2) gives $R_{h,p} \ge 0$. If this polynomial vanished identically, both F and $-F$ would have nonnegative real part in the right half-plane. The analytic function F would then be a purely imaginary constant. Its limit at infinity would force $F = 0$, contrary to $h \ne 0$. □

Let

$$J = \{\omega \in \mathbb{R} : h(i\omega) = 0\}, \qquad r = |J|, \qquad \ell = \deg h = n - 1. \tag{4.5}$$

Factor h over $\mathbb{C}$ into its imaginary-axis factor and a factor Q whose roots lie strictly in the left half-plane. Simplicity gives

$$\deg Q = \ell - r, \qquad \gcd(Q(s), h(-s)) = 1. \tag{4.6}$$

These statements use root-location stability directly; no coefficient version of the Routh–Hurwitz criterion is assumed in the formal main theorem.

5 Signed contacts and the reflection contradiction

5.1 Finite frequencies and moments

Split the atoms of (4.3) into finite atoms $F = \{j : |u_j| < 1\}$ and infinity atoms $|u_j| = 1$. For $j \in F$, set

$$\omega_j = \frac{u_j}{1 - |u_j|}, \qquad m_j = \mu_j (1 - |u_j|)^{2n-2} > 0. \tag{5.1}$$

There must be at least one finite atom. Indeed, at infinity the coordinate of Φ_p corresponding to s^{n-2} is -1, by (3.4) and monicity of p. If every atom were at infinity, that coordinate of the balance relation would read $-\sum_j \mu_j = 0$, contradicting $\sum_j \mu_j = 1$.

Introduce the symmetric set of distinct finite frequencies

$$S = \{\omega_j : j \in F\} \cup \{-\omega_j : j \in F\}, \qquad m = |S|. \tag{5.2}$$

For $w \in S$, define

$$\begin{aligned} \theta_{jw} &= \frac{m_j}{2}\big(\mathbf{1}_{\{\omega_j = w\}} + \mathbf{1}_{\{-\omega_j = w\}}\big), \\ r_w &= \sum_{j \in F} \theta_{jw}\, \overline{d_j(iw)}. \end{aligned} \tag{5.3}$$

This averaging treats a zero frequency correctly: its two half-weights add to its original weight. Conjugate symmetry and the contact equations imply

$$r_{-w} = \overline{r_w}, \qquad \operatorname{Re}\big(h(iw) r_w\big) = 0. \tag{5.4}$$

Every r_w is nonzero. To see this, group the weights at w by endpoint, obtaining $A, B \geq 0$, $A + B > 0$, and

$$r_w = \overline{Aa(iw) + Bb(iw)} = (A + B)\overline{p_{B/(A+B)}(iw)} \neq 0.$$

This is the place where stability of the full denominator segment ensures noncancellation of the grouped residues.

Set

$$M = \begin{cases} n - 1, & \text{if there is no infinity atom,} \\ n - 2, & \text{if there is an infinity atom.} \end{cases} \tag{5.5}$$

The balance relation yields the complex moment identities

$$\sum_{w \in S} r_w (iw)^k = 0 \qquad (0 \leq k < M). \tag{5.6}$$

Indeed, an infinity feature has nonzero coordinates only in the s^{n-2} and s^{n-1} directions. For every retained direction s^k, the finite part of the balance therefore gives

$$\sum_{j \in F} m_j \operatorname{Re}\big((i\omega_j)^k \overline{d_j(i\omega_j)}\big) = 0.$$

Pairing each frequency with its negative converts this real identity into (5.6). In the absence of infinity atoms it suffices to retain the first $n - 1$ directions; with infinity atoms, the first $n - 2$ directions remain available.

5.2 The contact-count bound

Let Z_a and Z_b be the sets of distinct real zeros of $R_{h,a}$ and $R_{h,b}$. Each $w \in S$ is a contact for at least one endpoint, so $S \subseteq Z_a \cup Z_b$. Every $w \in J$ is a zero of both cross polynomials, so $J \subseteq Z_a \cap Z_b$. Lemma 4.1 makes both cross polynomials nonzero and nonnegative. Each of their real zeros has multiplicity at least two. Hence

$$\begin{aligned} 2(m+r) &\le 2\big(|Z_a \cup Z_b| + |Z_a \cap Z_b|\big) \\ &= 2(|Z_a| + |Z_b|) \le \deg R_{h,a} + \deg R_{h,b}. \end{aligned} \tag{5.7}$$

This inequality counts both signs of every frequency and also the shared zeros contributed by h. It includes $w = 0$ without a separate convention.

If there is no infinity atom, (3.3) gives

$$m + r \le 2n - 2 = \ell + M. \tag{5.8}$$

If there is an infinity atom associated with endpoint p, its contact equation is $L_p(h) = 0$. The even polynomial $R_{h,p}$ then has degree at most $2n - 4$, while the other endpoint still has degree at most $2n - 2$. Therefore

$$m + r \le 2n - 3 = \ell + M. \tag{5.9}$$

The loss of one moment at infinity is thus accompanied by a reduction of one in the contact bound.

5.3 Residue reflection

Define the node polynomial and residue numerator by

$$D(s) = \prod_{w \in S} (s - iw), \qquad N(s) = \sum_{w \in S} r_w \frac{D(s)}{s - iw}. \tag{5.10}$$

Each quotient in this expression is a polynomial. At a node,

$$N(iw) = r_w D'(iw) \neq 0, \tag{5.11}$$

so N is nonzero. Expanding $N(s)/D(s) = \sum_w r_w/(s - iw)$ at infinity, the first M coefficients vanish by (5.6). Equivalently, comparison of polynomial coefficients gives

$$\deg N + M < m. \tag{5.12}$$

This includes $M = 0$, when it is just $\deg N < m$.

Lemma 5.1 (Reflection identity)**.** *The contact data satisfy*

$$h(s)N(s) = (-1)^m h(-s)N(-s). \tag{5.13}$$

Proof. Symmetry of S gives $D(-s) = (-1)^m D(s)$ and

$$D'(-iw) = (-1)^{m+1} D'(iw).$$

Since h has real coefficients, (5.4) implies

$$h(-iw)r_{-w} = \overline{h(iw)r_w} = -h(iw)r_w.$$

Together with (5.11), these identities show that

$$H(s) := h(s)N(s) - (-1)^m h(-s)N(-s)$$

vanishes at every iw, $w \in S$.

In both cases of (5.5), $\deg h = \ell \le M + 1$. Thus (5.12) gives $\deg(hN) \le m$. Even when equality occurs, the coefficient of s^m in H cancels:

$$[s^m]H = [s^m](hN)\big(1 - (-1)^m(-1)^m\big) = 0.$$

Consequently $\deg H < m$ unless $H = 0$. The m distinct zeros force $H = 0$, proving (5.13). □

The stable factor Q in (4.6) divides the left side of (5.13). Since it is coprime to $h(-s)$, it must divide $N(-s)$. Therefore

$$\ell - r = \deg Q \le \deg N \quad \Longrightarrow \quad \ell + M < m + r, \tag{5.14}$$

where the strict inequality follows from (5.12). This contradicts (5.8) or (5.9). No critical contact certificate can exist.

Proposition 5.2 (Compact lower-degree feasibility)**.** *For $n \ge 2$, if $p_\lambda \in \mathcal{H}_n$ for every $\lambda \in [0,1]$, there is a real polynomial c of degree at most $n-1$ such that*

$$B_a(c,u) > 0, \qquad B_b(c,u) > 0 \qquad (-1 \le u \le 1). \tag{5.15}$$

Proof. Otherwise Section 4 produces a critical certificate on a stable initial subsegment. The preceding contradiction excludes it. Equivalence (3.10) therefore supplies a strictly separating numerator. □

6 From region intersection to robust SPR synthesis

6.1 Necessity and extension to the entire segment

Suppose $a, b \in \mathcal{H}_n$ and one numerator c has strictly positive real-part quotients at both endpoints. Equations (3.2) and (3.5) imply

$$R_{c,p_\lambda}(\omega) > 0 \qquad (\lambda \in [0,1],\ \omega \in \mathbb{R}). \tag{6.1}$$

An imaginary-axis zero of p_λ would make the left-hand side zero. Thus the segment has no imaginary-axis roots.

For completeness, the stability continuation needed here can be established directly. Let E be the set of parameters at which p_λ has a root in the open right half-plane. The coefficients vary continuously on a compact interval, so a uniform Cauchy bound bounds every root. A convergent sequence of parameters in E has a convergent subsequence of such roots; the limiting root has nonnegative real part. Exclusion of imaginary-axis roots makes its real part positive. Hence E is closed.

The set E is also open. If $p_{\lambda_0}(z) = 0$ and $\operatorname{Re} z > 0$, choose $0 < \rho < \operatorname{Re} z$. For nearby λ, $|p_\lambda(z)| < \rho^n$. The factorization of the monic polynomial p_λ implies that some root lies within distance ρ of z, and consequently in the right half-plane. Since $[0,1]$ is connected and $0 \notin E$, the set E is empty. The whole segment is Hurwitz stable. Dividing (6.1) by $|p_\lambda(i\omega)|^2$ gives strict frequency positivity throughout the segment. The case $n = 0$ is immediate.

6.2 Exact lower degree and normalization

For $n \geq 2$, choose c from Proposition 5.2. Its finite-frequency quotients are strictly positive. In particular $c \neq 0$, and Lemma 4.1 gives $\deg c = n-1$. The same lemma excludes open right-half-plane zeros of c, and strict frequency positivity excludes imaginary-axis zeros. Hence c itself is Hurwitz stable.

Its leading coefficient is positive. Indeed, a monic real Hurwitz polynomial has positive constant term, so $a(0) > 0$. Evaluating (2.3) at zero gives $c(0) > 0$. A real polynomial all of whose roots lie in the open left half-plane has constant term with the same sign as its leading coefficient: negative real roots contribute positive factors, and conjugate pairs contribute positive products. Thus $\mathrm{lc}(c) > 0$. Replacing c by $c/\mathrm{lc}(c)$ preserves all strict inequalities and produces a monic numerator of degree exactly $n-1$.

If $n = 1$, write $a = s + \alpha$, $b = s + \beta$, with $\alpha, \beta > 0$. The common numerator $c = 1$ satisfies

$$\mathrm{Re}\,\frac{1}{i\omega + \alpha} = \frac{\alpha}{\alpha^2 + \omega^2} > 0, \qquad \mathrm{Re}\,\frac{1}{i\omega + \beta} = \frac{\beta}{\beta^2 + \omega^2} > 0.$$

This proves existence of a common monic lower-degree numerator for every $n \geq 1$. The endpoint extension above supplies statement 3 of Theorem 2.2, and necessity supplies its converse. In particular, it proves $\Omega_a^W \cap \Omega_b^W \neq \varnothing$ precisely when the denominator segment is stable.

6.3 A common degree-raising perturbation

The final step follows the perturbation principle of [1, 3].

Lemma 6.1 (Uniform perturbation)**.** *Let c be a common monic WSPR numerator for a stable segment of degree $n \geq 1$, and let g be any monic real polynomial of degree n. There is $\delta_0 > 0$ such that, whenever $0 < \delta \leq \delta_0$, the polynomial*

$$\widetilde{c}_\delta = c + \delta g \tag{6.2}$$

has degree n and leading coefficient δ, and $\widetilde{c}_\delta / p_\lambda$ is SPR for every $\lambda \in [0, 1]$.

Proof. For $p = a, b$, put $f_p = R_{c,p}$ and $g_p = R_{g,p}$. Then $f_p(\omega) > 0$ for every ω. The polynomial g_p is even, has degree $2n$, and has leading coefficient one. Choose $R > 0$ such that both g_a and g_b are positive for $|\omega| \geq R$. On $[-R, R]$, define

$$m_p = \min_{|\omega| \leq R} f_p(\omega) > 0, \qquad K_p = \max_{|\omega| \leq R} |g_p(\omega)|.$$

Any positive δ_0 satisfying

$$\delta_0 \leq \min\left\{ \frac{m_a}{2(1 + K_a)}, \frac{m_b}{2(1 + K_b)} \right\}$$

ensures $f_p + \delta g_p > 0$ on the compact interval for $0 < \delta \leq \delta_0$. Positivity outside that interval is automatic. Since

$$R_{c+\delta g, p} = f_p + \delta g_p,$$

both endpoint quotients are strictly frequency-positive. Affinity gives the result for every p_λ. The degree and leading coefficient follow from $\deg c = n - 1$ and monicity of g. □

Normalize by the same positive coefficient:

$$\beta_\delta = \delta^{-1}(c + \delta g) = g + \delta^{-1} c. \tag{6.3}$$

This is a monic polynomial of degree n, and positive rescaling preserves strict frequency positivity. Taking, for example, $g = a$ proves sufficiency in Theorem 2.3. Necessity and extension to the whole segment were established above. For $n = 0$, $a = b = \beta = 1$ completes the theorem.

The resulting argument has the single sequence

$$\begin{aligned} &\text{stable denominator segment} \implies \Omega_a^W \cap \Omega_b^W \neq \varnothing \\ &\implies \text{one monic degree-}(n-1)\text{ WSPR numerator} \\ &\quad\implies \text{one monic degree-}n\text{ SPR numerator.} \end{aligned} \tag{6.4}$$

Both numerators are independent of the segment parameter and of the frequency. The degree bounds are exact, and there is no upper restriction on n.

7 Conclusion

The WSPR region-intersection method gives a unified route to robust SPR synthesis for polynomial segments. The formalization establishes intersection in every positive order, normalizes the common numerator to exact degree $n - 1$, and derives the same-degree result by a single perturbation. The central exclusion argument compares a contact-count bound with a degree inequality forced by vanishing residue moments and reflection. Contacts at infinity reduce both the moment count and the admissible contact count, so the same contradiction remains valid. This supplies a machine-checked algebraic realization of the established geometric synthesis strategy, including the necessity direction and one fixed numerator for the entire segment.

8 Formal verification

The proofs presented in this paper have been formally verified in Lean. The project repository is `https://github.com/XPiPiT/n_random/tree/main/SPR/simplified`.

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